Design And Analysis of Algorithms
Lecture 13: Multiplicative Weights, Zero-Sum Games and Solving LPs

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Last time, we designed algorithms which learn from expert advice: given a sequence of losses for each expert, the algorithm automatically adjusts to play mixtures over those experts which nearly matches the loss of the best fixed expert in hindsight. Today, we'll use the algorithm we designed to solve zero-sum games and LPs.

Everything can be phrased either in terms of losses, which we're trying to minimize, or gains, which we're trying to maximize. It's standard to consider gains for games and LPs, though the problems are equivalent.

Theorem 1. *Suppose loss vectors $\ell^t \in [-\gamma, \rho]^N$. Then for any $\epsilon < \frac{1}{2}$, and for any $T \geq \frac{4\gamma\rho \ln N}{\epsilon^2}$, The hedge algorithm outputs $p^1, \ldots, p^T \in \Delta_N$ such that for any $i \in [N]$,*

$$\frac{1}{T}\sum_{t=1}^{T}\langle p^t, \ell^t\rangle \leq \frac{1}{T}\sum_{t\leq T}\ell_i^t + \epsilon.$$

Equivalently, if we use gain vectors rather than loss vectors, and Hedge is maximizing rather than minimizing, we have

$$\frac{1}{T}\sum_{t=1}^{T}\langle p^t, \ell^t\rangle \geq \frac{1}{T}\sum_{t\leq T}\ell_i^t - \epsilon.$$

1 Zero-Sum Games

A zero-sum game is defined by a matrix $M \in \mathbb{R}^{m\times n}$, where there is a "row" player who picks a row $r \in [m]$ and a "column" player who picks a column $c \in [n]$. Fixing both actions, the value $M_{r,c}$ is called the payoff of the row player; we assume the column player receives $-M_{r,c}$, thus the name "zero-sum" since the payoffs sum to zero. Similarly, for $x \in \Delta_m, y \in \Delta_n$, the payoff of the row player is $x^T My$, which is the negative of the payoff for the column player.

If the two players receive these payoffs, it is natural for both of them to maximize the payoff they get. Formally, fixing some strategy for the row player $x \in \Delta_m$, y would ideally like to maximize her payoff (or, equivalently, minimize the row player's payoff). That means she'd be able to pick a column which give her payoff

$$C(x) = \min_y x^T My = \min_{j\in[n]} x^t Me_j$$

and, respectively, fixing $y \in \Delta_n$, the row player will be able to pick a row such that he guarantees himself of

$$R(y) = \max_x x^T My = \max_{i\in m} e_i My.$$

The row player wants to pick an x to maximize $C(x)$, and the column player to pick a y which minimizes $R(y)$.

Remark 1. *For any $x \in \Delta_m, y \in \Delta_n$, we have that $C(x) \leq R(y)$, since*

$$C(x) = \min_{\eta} x^T M\eta \leq \max_{z} \min_{\eta} z^T M\eta \leq \max_{z} z^T My = R(y).$$

Moreover, maximizing the left-hand side and minimizing the right-hand side over x and y respectively gives the same value. In other words, the row player choosing an action which maximizes their minimum payoff and then the column player best-responding gives the same payoff as the column player first choosing an action which minimizes the row player's maximum payoff and the row player best-responding.

Theorem 2 (Von-Neumanns' Minimax)**.** *For any finite $M \in \mathbb{R}^{m\times n}$,*

$$\max_{x\in\Delta_m} C(x) = \min_{y\in\Delta_n} R(y).$$

Proof. Remark 1 implies $C(x) \leq R(y)$ for all x, y, so assume for contradiction there exists some $\delta > 0$ such that

$$\max_{x\in\Delta m} C(x) < \min_{y\in\Delta_n} R(y) - \delta.$$

We will show we can use the Hedge algorithm to update the row's action to get better payoff and arrive at a contradiction. Let $p^1 = (\frac{1}{m}, \ldots, \frac{1}{m})$, the row player will play p^t for each round t. Assume the column player best-responds to p^t, meaning they pick the action which maximizes their reward,

$$j^t = \arg\min_{j\in[n]} (p^t)^T Me_j.$$

Then, the payoff for choosing each action $i \in [n]$ for the row player fixing the column player's behavior is

$$g^t = Me_{j^t}$$

which will be used to update the Hedge algorithm. Then, let

$$\hat{x} = \frac{1}{T}\sum_t p^t \qquad \text{and} \qquad \hat{y} = \frac{1}{T}\sum_t e_{j^t}$$

By Theorem 1, we know that the average row player's behavior is nearly optimal, i.e. that

$$C(\hat{x}) = \frac{1}{T}\sum_t \langle p^t, g^t\rangle \geq \max_i \frac{1}{T}\sum_i \langle e_i, g^t\rangle, -\epsilon = R(\hat{y}) - \epsilon.$$

□

Moreover, our remark implies $C(\hat{x}) \leq R(\hat{y})$; since ϵ is under our control merely as a function of the number of rounds we run Hedge, we can drive it to being less than δ, a contradiction.

2 Solving Linear Programs

We now show how to use Hedge to (approximately) solve a linear program:

$$\begin{array}{lll} \text{maximize} & c^T x & \\ \text{subject to} & Ax \leq b & \\ & x_i \geq 0, & \forall i. \end{array}$$

The idea will be to update hedge according to penalties which are larger for constraints which we violate more severely. Define $F = \max_{x \in \text{feasible solutions}} c^T x$, the optimal feasible value of the LP. Let the set of possible optimal points for our LP be defined as

$$K = \{x : x \geq 0, c^T x = F\}$$

We first observe that if we only have a single constraint $\alpha x \leq \beta$, we can quickly find a point which satisfies that constraint and achieves the correct value, or outputs that the constraint is infeasible for that objective value.

Observation 1. *For $\alpha, c \in \mathbb{R}^n, \beta \in \mathbb{R}$, and some value F, there is an efficient Oracle that either finds x^* for which $c^T x^* = F$ and $\alpha x^* \leq \beta$, or output that no such x^* exists.*

Proof. Assume $M \in [-1, 1]^{m \times n}, c_i > 0$ for all i (the other proof is analogous.) In order to maximize $c^T x$ subject to the constraint, one should order the coordinates by their "bang per buck" (the amount of objective gained versus the tightening of the constraint). The coordinate that earns the most objective per unit of constraint, $j^* = \arg\max_j \frac{C_j}{\alpha_j}$, needs to net us sufficient objective value, i.e.

$$x = e_{j^*} \cdot \min\left\{\frac{F}{c_{j^*}}, \frac{\beta}{\alpha_{j^*}}\right\}$$

and, if $c^T x = F$, then return x (as it is feasible and achieves the desired objective); if not, return "infeasible". This latter return is justified, since row j^* will be optimal for maximizing the objective subject to the constraint. □

So, our algorithm will use Hedge to produce new weightings over the constraints which aren't satisfied, and we will then call this oracle to find an x^t which maximizes the objective value subject to that convex combination of constraints.

Algorithm 1: Hedge-LP(F, A, b, c)

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Let p^1 = (1/m, ... 1/m);
for t = 1 to T ≥ 4ργ ln N / ε^2 do
    Call Oracle to find x^t which satisfies ⟨Σ_i a_i p_i^t, x^t⟩ ≤ Σ_i b_i p_i^t with value F;
    if The oracle returns infeasible then
        return infeasible;
    end
    else
        Let ℓ_i^t = b_i − (Ax^t)_i;
        Let p^t = Hedge(ε, ℓ^1, ..., ℓ^t);
    end
end
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Theorem 3. *Hedge-LP produces a solution for which $\max_i |a_i x - b| \leq \epsilon$ with value F if such an x exists, and outputs "infeasible" otherwise.*

Proof. Since every $x^t \in K$, $\hat{x} \in K$, so the value of the solution returned is optimal. Moreover, since the oracle is given convex combinations of the constraints, if it returns infeasible the LP is infeasible (any convex combination of the constraints should be satisfiable if the LP is satisfiable).

So, it remains to show that for any i, $|a_i x - b_i| \leq \epsilon$. Theorem 1 implies that

$$\frac{1}{T}\sum_t \langle p^t, \ell^t \rangle \leq \frac{1}{T}\sum_t \langle e_i, \ell^t \rangle + \epsilon = \sum_t (b - Ax^t)_i + \epsilon$$

And we know by definition that

$$\langle p^t, \ell^t \rangle = \langle p^t, b - Ax^t \rangle = \langle p^t, b \rangle - \langle p^t, Ax^t \rangle = \beta^t - \alpha^t x^t \geq 0$$

where the last inequality follows from the fact that our oracle gauranteed this constraint was satisfied. Combining these two, we have that $\sum_t (b - Ax^t)_i \geq -\epsilon$ as desired.

$\square$